INDUSTRIAL DIAGNOSTIC & INVESTIGATION REPORT

HIGH SEVERITY

Sovereign Multi-Agent AI Workbench — Plant Intelligence System

Report ID:	`REP-20260831-5346B2`	**Equipment ID:**	**Pump P-101**
Plant Unit / Area:	FCCU Crude Distillation Unit 2	**Report Standard:**	ISO 55001 / OSHA 1910
Report Type:	Incident Investigation	**Status:**	**APPROVED / ACTIVE**

1. Executive Summary

This Incident Investigation was conducted for Pump P-101 located in FCCU Crude Distillation Unit 2. Based on aggregated diagnostic evidence (Severity: HIGH), operational anomalies and surface/procedural conditions have been documented. Immediate corrective actions and scheduled preventative inspections are detailed herein.

2. Problem Statement & Scope

Investigation and diagnostic synthesis conducted for task: **Centrifugal Pump P-101 Severe Vibration & Flange Corrosion Investigation**. Multi-agent cross-referencing was deployed to correlate direct visual indicators, sensor telemetry trends, and documented standard operating procedures (SOPs).

3. Root Cause Analysis (5-Whys Methodology)

Step 1 Why: An operational anomaly was detected on Pump P-10

Step 2 Why: Physical inspection or telemetry indicated deviations from standard baseline parameters.

Step 3 Why: Component wear, seal degradation, or operational stress exceeded nominal tolerances.

Step 4 Why: Scheduled preventive maintenance interval or operating condition thresholds were reached.

Step 5 Root Cause: Environmental/operational stress coupled with regular duty-cycle wear requiring overhaul.

4. Actionable Corrective & Preventative Recommendations

1. **Action 1:** Perform immediate physical lockdown and ultrasonic verification on Pump P-101.
2. **Action 2:** Execute standard operating procedure compliance check against latest plant SOP revisions.
3. **Action 3:** Replace worn seals, re-torque flange bolts to specified Nm rating, and conduct pressure test.
4. **Action 4:** Log diagnostic findings in the Plant Maintenance ERP for ongoing condition monitoring.

5. Traceability & Evidence Citations Appendix

#	Source Document / Image	Reference Chunk / Observation	Confidence
1	**pump_p101_flange.jpg**	Bolts 2 & 4 rust visible, gauge at 1.4 bar	92%
2	**MRPL-SOP-PUMP-2024.pdf**	Section 4.2 Minimum Pressure Tolerances	95%

3	**telemetry_p101.csv**	Timestamp 2026-08-30 08:30 vibration anomaly	**90%**

Lead Plant Engineer	**Operations Superintendent**	**Safety & Compliance Officer**
Inspection Lead	Shift In-Charge	ISO / HSE Auditor